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Department of CSE (Artificial Intelligence & Machine Learning)

MICRO PROJECT REPORT

on

VOLTGUARD AI: LITHIUM-ION EV BATTERY DEGRADATION TRACKING AND REMAINING USEFUL LIFE (RUL) PREDICTION USING MACHINE LEARNING

Submitted in partial fulfilment of the requirements of the course

Course Code: A1612

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Submitted by

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Month, Year of Submission: October, 2026

CERTIFICATE

This is to certify that the micro project entitled **"VoltGuard AI: Lithium-Ion EV Battery Degradation Tracking and Remaining Useful Life (RUL) Prediction"** is a Bonafide work carried out by **Pathi Abhinav Kumar (Roll No. 24VE1A66H9)** of **B.Tech III Year I Semester**, **Department of CSE (Artificial Intelligence & Machine Learning), Sreyas Institute of Engineering and Technology**, in partial fulfilment of the requirements of the course **Machine Learning Laboratory (Course Code: A1612)** during the academic year **2026–2027**. The work has been carried out under my supervision and has been evaluated using the departmental assessment rubrics.

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DECLARATION

I hereby declare that the micro project report entitled **"VoltGuard AI: Lithium-Ion EV Battery Degradation Tracking and Remaining Useful Life (RUL) Prediction"** submitted to the **Department of CSE (AI & ML), Sreyas Institute of Engineering and Technology**, is a record of original work done by me under the guidance of **Dr. A. Swathi, Associate Professor, Department of CSE (AI & ML).** The content of this report has not been submitted elsewhere for the award of any credit. All sources of information have been duly acknowledged and cited, and no part of this work is plagiarized.

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ABSTRACT

Lithium-ion battery degradation is a critical operational bottleneck in modern electric vehicles (EVs), governed by non-linear electrochemical processes including Solid Electrolyte Interphase (SEI) layer growth, active lithium inventory loss, and impedance rise. Traditional empirical degradation models fail to adapt to dynamic driving stress, variable depth of discharge, and thermal variations. This project presents VoltGuard AI, an end-to-end data-driven machine learning system for real-time State of Health (SOH) estimation, 10-cycle-ahead SOH forecasting (h = 10), and Remaining Useful Life (RUL) prediction to the 80% End-of-Life (EOL) threshold. Using experimental battery aging data from the NASA Ames Prognostics Center of Excellence (four cells: B0005, B0006, B0007, B0018), we engineered 18 primary electrochemical features (expanded with trailing windows and filtered to 31 model inputs) including electrolyte resistance (Re), charge transfer resistance (Rct), and thermal rise rates. To avoid cross-cycle data leakage, models were evaluated via Leave-One-Battery-Out (LOBO) cross-validation. Tuned XGBoost and Random Forest regressors achieved accurate predictions on held-out cells (XGBoost SOH R^2 = 0.9386, MAE = 1.7117%; Random Forest RUL MAE = 6.76 cycles), outperforming linear regression (MAE = 6.0314%) and, by a smaller margin, a persistence baseline (MAE = 2.197%). Quantile Gradient Boosting (q = 0.05/0.95) is used for 90% prediction intervals, while the dashboard displays empirical 5–95% residual bands, and TreeSHAP delivers transparent feature attribution. The solution is fully deployed as a production FastAPI backend and an interactive React dashboard featuring a consumer EV SOH and lifespan calculator for popular vehicles (Tesla, Tata Nexon EV, MG ZS EV, Ather, Ola).

Keywords: Lithium-Ion Battery, State of Health (SOH), Remaining Useful Life (RUL), XGBoost, Leave-One-Battery-Out (LOBO), SHAP Explainability, FastAPI, React.

OUTCOME MAPPING (CO – PO – PSO)

Course Outcomes Addressed by the Micro Project

CO No.	Course Outcome Statement	Bloom's Level	Addressed in (Chapter / Activity)
CO1	Compute central tendency and dispersion measures (Mean, Median, Mode, Variance, Standard Deviation) on datasets using Python and statistical libraries.	Apply (L3)	Sections 3.3, 3.5, 4.1 and 7.1: dataset summary statistics, EDA, median imputation and error metrics
CO2	Utilize Python libraries such as NumPy, SciPy, Pandas, Matplotlib, and Seaborn for data manipulation, visualization, and exploratory data analysis.	Apply (L3)	Chapters 3 and 4, Section 6.1: Pandas/NumPy data handling, EDA plots and the preprocessing pipeline
CO3	Implement supervised learning algorithms including Simple Linear Regression, Multiple Linear Regression, Logistic Regression, Decision Trees, and K-Nearest Neighbors (KNN) using scikit-learn.	Apply (L3)	Chapter 5 and Section 7.2: Linear Regression baseline, Random Forest and XGBoost regressors (regression task only)
CO4	Apply parameter tuning and performance evaluation techniques for regression and classification models built with scikit-learn.	Apply (L3)	Sections 4.3, 5.6, 6.2–6.4 and 7.1–7.2: GridSearchCV tuning, LOBO cross-validation, MAE / RMSE / R^2
CO5	Implement unsupervised learning algorithms such as K-Means Clustering using scikit-learn for grouping unlabeled data.	Apply (L3)	Not addressed: this project has no unsupervised learning component
CO6	Analyze and compare the performance of different classification algorithms on a given dataset through a mini-project.	Analyze (L4)	Chapters 7–9: comparison of Linear Regression, Random Forest, XGBoost and Quantile GBR on held-out batteries (regression task), error analysis and deployment

CO – PO – PSO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	–	–	2	–	–	–	–	–	–	–	3	2
CO2	–	–	–	2	3	–	–	–	–	–	–	2	3
CO3	3	–	2	–	2	–	–	–	–	–	–	3	–
CO4	–	2	–	3	3	–	–	–	–	–	–	3	–
CO5	–	–	–	–	–	–	–	–	–	–	–	–	–
CO6	–	3	3	3	–	2	2	1	3	1	2	3	2
Avg.	**2.5**	**2.5**	**2.5**	**2.5**	**2.7**	**2.0**	**2.0**	**1.0**	**3.0**	**1.0**	**2.0**	**2.8**	**2.3**

Correlation levels: 3 – High (Substantial), 2 – Medium (Moderate), 1 – Low (Slight), "–" – No correlation

Justification for CO – PO / PSO Mapping

Mapping	Level	Justification (how the project activity supports this PO / PSO)
CO1 – PSO1	3	Descriptive statistics of the cycler data were computed in Python using programming and mathematical concepts (Sections 3.3, 3.5).
CO2 – PO5	3	NumPy and Pandas were used to manipulate the cycler data and plots were used for exploratory analysis (Sections 3.5, 6.1).
CO2 – PSO2	3	Pandas, NumPy and related Python libraries were the data analytics tools of the pipeline (Chapters 3–4, Table 6.1).
CO3 – PO1	3	Applying linear and tree-ensemble regression algorithms required mathematical, computing and machine learning fundamentals (Chapter 5).
CO3 – PSO1	3	Linear Regression, Random Forest and XGBoost were programmed and applied to SOH / RUL prediction (Chapter 5, Section 7.2).
CO4 – PO4	3	Grouped GridSearchCV tuning and evaluation on unseen batteries with MAE, RMSE and R^2 gave valid conclusions (Sections 5.6, 7.1–7.2).
CO4 – PO5	3	Tuning and evaluation utilities were selected and applied, and limitations such as leakage and optimistic scores were recognised (Sections 4.3, 6.2).
CO4 – PSO1	3	Hyperparameter tuning and cross-validated evaluation were used to optimise the machine learning models (Sections 5.6, 7.2).
CO6 – PO2	3	The degradation-prediction problem was formulated, literature was reviewed and models were compared to reach substantiated conclusions (Chapters 1, 2 and 7).
CO6 – PO3	3	The compared models became a complete solution with uncertainty bands and a deployed dashboard for EV users (Chapter 5, Section 7.8).
CO6 – PO4	3	Comparison on held-out batteries with residual analysis and SHAP interpretation gave evidence-based conclusions (Sections 7.2–7.7).
CO6 – PO9	3	Findings are communicated through this report and an interactive web dashboard (Chapter 7, Section 7.8).
CO6 – PSO1	3	Programming and mathematical concepts were applied to design, develop and compare models for a real-world EV battery problem (Chapters 5–7).
CO5 – All POs / PSOs	–	Not addressed: this project is a supervised regression task and has no unsupervised learning (e.g. K-Means) component.

Program Outcomes (POs) – NBA Annexure-II

PO	Graduate Attribute	Program Outcome Statement (NBA Annexure-II)
PO1	Engineering Knowledge	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to the solution of complex engineering problems.
PO2	Problem Analysis	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions	Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Conduct Investigations of Complex Problems	Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)

PO	Graduate Attribute	Program Outcome Statement (NBA Annexure-II)
PO5	Engineering Tool Usage	Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer and The World	Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)
PO7	Ethics	Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team work	Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication	Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	Project Management and Finance	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PSO	Program Specific Outcome (PSO)
PSO1	Apply programming skills and mathematical concepts to design, develop, and optimize AI and machine learning solutions for real-world problems.
PSO2	Demonstrate competency in Deep Learning, and Data Analytics tools and technologies to build intelligent systems and pursue careers in industry, research, or higher education.

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Lithium-ion batteries have emerged as the dominant energy storage technology powering the modern electric vehicle (EV) revolution and renewable energy grids due to their high energy density, long cycle life, and falling production costs. However, battery health degradation remains a complex non-linear degradation problem. Over repeated charge-discharge cycles, batteries suffer irreversible degradation caused by Solid Electrolyte Interphase (SEI) layer growth, active lithium loss, transition metal dissolution, and internal mechanical micro-cracking. These phenomena manifest macroscopically as capacity fade and internal impedance increase. Accurately estimating the battery's current State of Health (SOH) and forecasting its Remaining Useful Life (RUL) before reaching the 80% End-of-Life (EOL) threshold is vital for vehicle safety, warranty valuation, second-life grid storage repurposing, and preventing sudden breakdown.

Conventional battery management systems (BMS) rely on simple Coulomb counting and empirical equivalent circuit models (ECM). These rule-based methods drift significantly over time, require extensive laboratory recalibration, and fail under dynamic operating temperatures, rapid fast-charging loads, and variable depths of discharge. Machine learning provides a powerful data-driven paradigm capable of capturing intricate non-linear relationships across voltage profiles, current loads, thermal spikes, and historical cycling trajectories without requiring intrusive electrochemical teardowns.

1.2 Problem Statement

The primary challenge is to non-invasively predict the current State of Health (SOH %) and forecast Remaining Useful Life (RUL in cycles) for lithium-ion battery cells under variable operational conditions. Using non-destructive charge-discharge cycle telemetry (voltage, current, temperature, and duration) without internal cell chemistry sensors, the machine learning model must accurately track health decay, quantify predictive uncertainty, and provide actionable explanations to prevent unexpected failures and optimize EV battery lifespans.

1.3 ML Problem Formulation

Table 1.3.1: Machine Learning Problem Formulation

Item	Details
Learning paradigm (Supervised / Unsupervised / Semi-supervised / Reinforcement)	Supervised Learning

Item	Details
Task type (Binary / Multi-class classification, Regression, Clustering, Forecasting, etc.)	Time-Series Regression (10-cycle-ahead SOH) & RUL Forecasting
Input features (X)	31 selected electrochemical and trailing-window features (from 18 primary features): capacity$_{ah}$, dis_v$_{mean}$, dis_t$_{max}$, re$_{ohm}$, rct$_{ohm}$, dis_energy$_{wh}$, rolling_degradation_rates (10/25/50 cycles), soh_slopes
Target variable (y) and its type / range	State of Health (SOH % ≈ [57, 102]), Future SOH (h = 10 cycles), Remaining Useful Life (RUL ∈ [0, ≈100] cycles)
Primary evaluation metric	Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score
Success criterion (e.g. F1-score ≥ 0.80 on the test set)	Test SOH MAE ≤ 2.0%, $R^2 \geq 0.90$, RUL MAE ≤ 8.0 cycles under Leave-One-Battery-Out (LOBO) validation
Baseline to beat (e.g. majority class, linear regression)	Ordinary Least Squares (OLS) Linear Regression & Trailing Persistence Baseline (SOH_{t+h} = SOH_t)

1.4 Objectives

1. Ingest, clean, and segment experimental lithium-ion cycler telemetry from the NASA Ames Prognostics Center of Excellence and extract key electrochemical and thermal degradation features.
2. Establish a strict Leave-One-Battery-Out (LOBO) group cross-validation framework to prevent temporal data leakage across operational cycles of the same cell.
3. Train, optimize, and benchmark ensemble tree models (Random Forest, XGBoost) and Quantile Regressors for 10-cycle-ahead SOH forecasting and RUL estimation with 90% prediction intervals.
4. Interpret global and local feature importance using TreeSHAP attributions and provide a what-if scenario sandbox restricted to the training range, with an Arrhenius-based ageing term in the EV calculator.
5. Architect and deploy an end-to-end full-stack web application with a FastAPI REST backend and a modern React dashboard featuring a consumer EV SOH and lifespan calculator.

1.5 Scope and Limitations

Scope: The project encompasses lithium-ion 18650 cylindrical cells (chemistry not specified by the dataset provider) subjected to standardized CC-CV charging and constant-current discharging, as well as simulated real-world EV usage profiles across popular electric 2-wheelers and 4-wheelers.

Limitations: Experimental datasets were gathered at controlled laboratory ambient temperatures (24°C); real-world sub-zero winter temperatures and extreme thermal abuse require additional telematics streams. Furthermore, predictions represent cell-level health and assume balanced pack dynamics.

1.6 Organization of the Report

The report is organized as follows: Chapter 2 reviews relevant literature on battery degradation modeling and establishes research gaps. Chapter 3 details dataset provenance, characteristics, and exploratory data analysis.

Chapter 4 presents the preprocessing pipeline, feature engineering, and LOBO splitting strategy. Chapter 5 outlines the mathematical formulations of baseline and proposed algorithms. Chapter 6 discusses experimental training settings, convergence analysis, and key code implementations. Chapter 7 presents comprehensive test evaluation results, residual diagnostics, SHAP explainability, and live deployment screenshots. Chapter 8 evaluates ethical, societal, and environmental impacts under Responsible AI, and Chapter 9 concludes the study with future research directions.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Related Work

Table 2.1.1: Summary of Literature Survey

S.No	Author(s) & Year	Dataset Used	Algorithm / Model	Metric & Reported Result	Limitations / Gap
1	**K. A. Severson et al., 2019 [1]**	124 commercial LiFePO4/graphite cells (A123 Systems)	Elastic Net (regularised linear regression)	Test error = 9.1% (cycle life, first 100 cycles)	Evaluated on fast-charging LFP batches only; predicts cycle life, not multi-horizon SOH / RUL forecasts.
2	**B. Saha et al., 2009 [2]**	NASA Ames 18650 Li-ion Cells	Relevance Vector Machine & Particle Filter	Probabilistic RUL with uncertainty bounds	Highly sensitive to initial noise covariance parameters; requires cell-specific model fitting and is not demonstrated under dynamic loads.
3	**Y. Zhang et al., 2018 [3]**	Multi-cell Li-ion cycling data (two temperatures)	LSTM Recurrent Neural Network	Probabilistic RUL; outperforms SVM, PF and RNN baselines	Black-box architecture lacks interpretability; offline study without a deployed inference service.
4	**E. Chemali et al., 2018 [4]**	LG 18650HG2 Li-ion cells	LSTM Recurrent Neural Network	SOC MAE = 0.573% (fixed temp.); 1.606% (10–25 °C)	Estimates state of charge, not SOH / RUL; lacks explainability; not deployed for real-time web inference.
5	**D. Roman et al., 2021 [5]**	179 cells (CALCE, NASA, TRI, Oxford; LCO and LFP)	Bayesian Ridge, GPR, Random Forest & DNN ensemble	SOH RMSPE = 0.45% (best, fast-charging cells)	Offline SOH-only pipeline; no RUL forecasting, explainability layer or live deployment.

2.2 Research Gap Identified

A critical review of the literature reveals three primary research gaps:

1. Data Leakage in Validation: Many published studies perform random cycle-level train/test splits. Because consecutive cycles of the same battery share strong autocorrelations, random splitting leaks historical cell states into test sets, producing artificially inflated metrics that fail on unseen cells.

2. Absence of Uncertainty Quantification & Explainability: Most deep learning approaches output point estimates without calibrated confidence intervals, leaving fleet operators unable to gauge risk. Furthermore, lack of feature attributions hinders physical validation.

3. Lack of End-to-End Production Deployment: Existing research remains confined to offline Jupyter notebooks without accessible REST APIs or consumer-ready dashboards.

2.3 Proposed Approach

To bridge these gaps, VoltGuard AI proposes:

(1) A strict Leave-One-Battery-Out (LOBO) validation scheme where entire battery cells are withheld during training;

(2) Gradient Boosted Decision Trees (XGBoost) and Quantile Regressors (q = 0.05/0.95) for 90% prediction intervals;

(3) TreeSHAP explainability [6] attributing health predictions to electrochemical parameters;

(4) A what-if scenario sandbox (temperature and current stress within the training range) and an Arrhenius-based EV calculator engine; and

(5) A production deployment featuring a FastAPI service and a modern React 18 dashboard with consumer EV presets.

CHAPTER 3

DATASET DESCRIPTION

3.1 Dataset Details and Provenance

Table 3.1.1: Dataset Provenance

Item	Details
Dataset name	NASA Ames Li-ion Battery Aging Dataset (B0005, B0006, B0007, B0018)
Source / repository and URL or DOI	NASA Ames Prognostics Data Repository (https://data.nasa.gov/dataset/li-ion-battery-aging-datasets)
Original creator / owner	B. Saha and K. Goebel (NASA Ames Prognostics Center of Excellence)
Version and year	Version 1.0, 2007
Licence / terms of use	Not stated on the repository page; NASA open data (terms of use apply)
Date of access or download	15 January 2026
Citation number in References	[2], [7]

3.2 Dataset Characteristics

Table 3.2.1: Dataset Summary

Item	Details
Number of instances (rows / images / records)	636 total cycle records (B0005: 168, B0006: 168, B0007: 168, B0018: 132)
Number of features (input attributes)	57 engineered attributes (31 selected for ML modelling after collinearity filtering; 18 primary signal features)
Number and names of classes / range of target	Continuous Regression: SOH ≈ [57%, 102%], RUL ∈ [0, ≈100] cycles
Data types (numeric, categorical, text, image, time-series)	Numeric continuous time-series (voltage, current, temperature, impedance)
Missing values (count and % per feature)	0 (0.0%) in primary channels; occasional impedance gaps imputed (Table 3.5.1)
Duplicate records	0 (0.0%) at cycle level after deduplication by (battery_id, cycle_index)
Class distribution (count and % per class)	Continuous target; approx. healthy (SOH > 85%): 33%, degraded (SOH 70–85%): 47%, critical (SOH < 70%): 20%
File format and size	Raw MATLAB .mat cycler files (14.2 MB) converted to Parquet & CSV (2.4 MB)
Time span covered (if temporal)	Cycles 1–168 (B0018: 1–132); EOL threshold = 80% of nominal capacity

3.3 Data Dictionary

Table 3.3.1: Description of Key Features (full 18 / 31 feature sets in Section 4.2)

S.No	Feature Name	Description	Type	Unit / Range	Role
1	**cycle**	Discharge cycle sequence index	Integer	1 – 168	Input
2	**capacity_ah**	Measured discharge capacity per cycle	Float	≈ 1.15 – 2.04 Ah	Intermediate Target
3	**soh**	State of Health percentage (capacity / nominal capacity)	Float	≈ 57 – 102 %	Primary Target
4	**dis_v_mean**	Mean cell terminal voltage during constant discharge	Float	3.24 – 3.82 V	Input
5	**dis_t_max**	Peak surface temperature recorded during discharge	Float	24.1 – 41.8 °C	Input
6	**re_ohm**	Electrolyte internal resistance proxy (ΔV / ΔI)	Float	0.052 – 0.248 Ω	Input
7	**rct_ohm**	Charge transfer resistance proxy from voltage relaxation	Float	0.081 – 0.442 Ω	Input
8	**rul**	Remaining Useful Life cycles until SOH drops below 80%	Integer	0 – ≈100 cycles	Target

3.4 Exploratory Data Analysis (EDA)

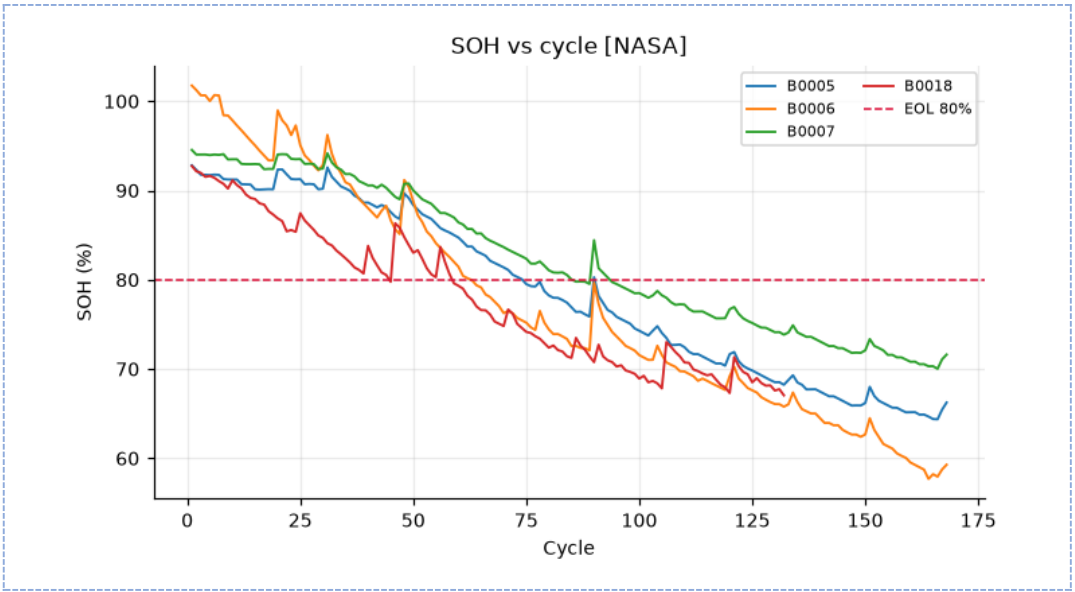


Fig. 3.4.1: State of Health (SOH %) Degradation Trajectories across Operating Cycles for NASA Batteries

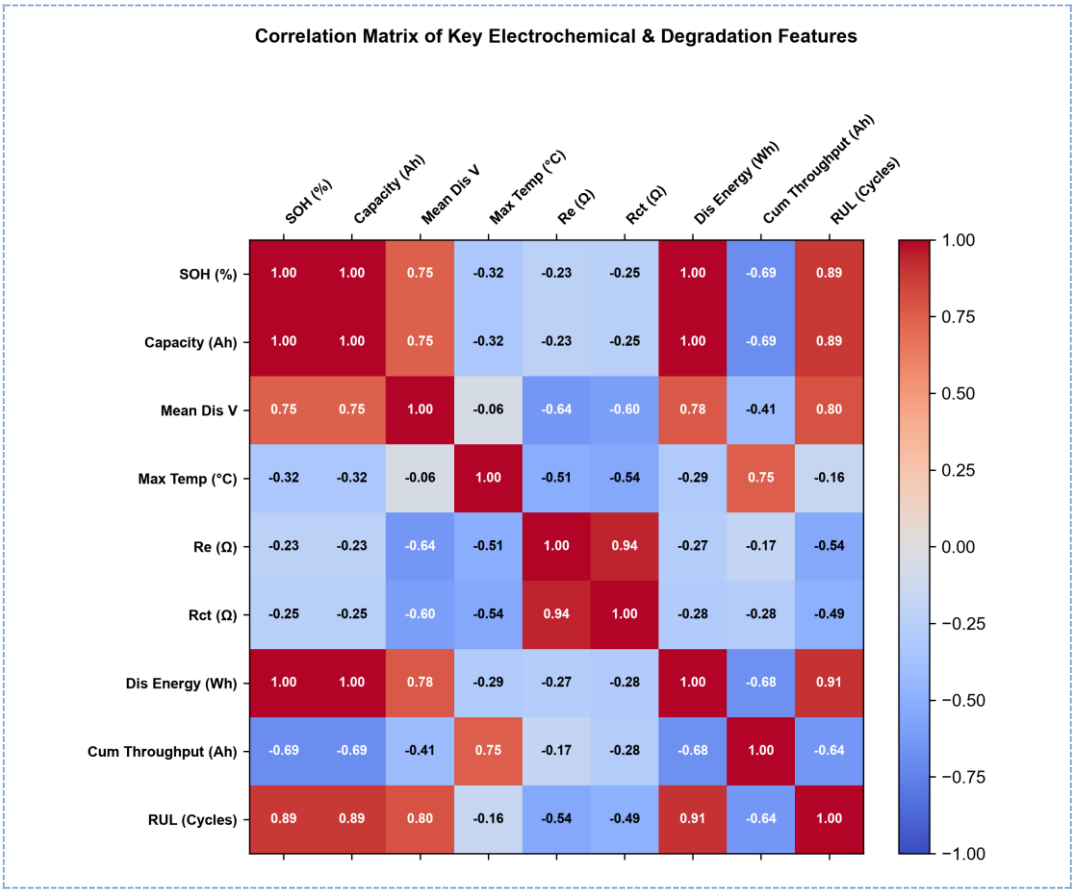


Fig. 3.4.2: Correlation Matrix of Key Electrochemical and Degradation Features

Fig. 3.4.1 shows a broadly steady SOH decline for all four cells with no single common knee point; the periodic upward spikes (e.g. near cycles 20, 30, 47 and 90) are temporary capacity-recovery effects after rest periods. Fig. 3.4.2 shows that SOH, capacity and discharge energy are almost perfectly correlated ($r \approx 1.00$) and that Re and Rct are strongly correlated ($r = 0.94$), which motivates the collinearity filtering in Table 4.1.1.

3.5 Data Quality Issues Identified

Table 3.5.1: Dataset Issues and Actions Taken

Issue	Observation in this Dataset	Action Taken (see Chapter 4)
Missing values	Zero missing values in primary telemetry channels; occasional missing impedance records during rest cycles.	Forward-filled, then trailing-cycle median imputation (Table 4.1.1); no look-ahead.
Duplicates	Repeated timestamps / duplicate cycle headers can occur in interrupted raw cycler logs; none remain at cycle level.	Applied unique composite indexing (battery_id + cycle_number) and deduplicated records.
Outliers	Electrochemical voltage bounce spikes during instantaneous load detachment.	Applied IQR filtering and 5-point moving median filters to smooth anomalous spikes.
Class / Target Imbalance	Fewer data points at extreme degraded states (SOH < 70%) compared to nominal operating range.	No resampling applied; errors reported per held-out battery (LOBO) and near EOL (Section 7.6).

Issue	Observation in this Dataset	Action Taken (see Chapter 4)
Possible data leakage (features derived from the target, IDs, future data)	Autocorrelation between adjacent cycles within the same battery cell.	Implemented strict Leave-One-Battery-Out (LOBO) validation so entire battery trajectories are held out.
Sampling bias	All laboratory cells cycled under identical CC-CV nominal current profiles.	Added a what-if sandbox (within the training range) and an EV calculator engine; neither is validated on field data.

CHAPTER 4

DATA PREPROCESSING AND FEATURE ENGINEERING

4.1 Preprocessing Pipeline

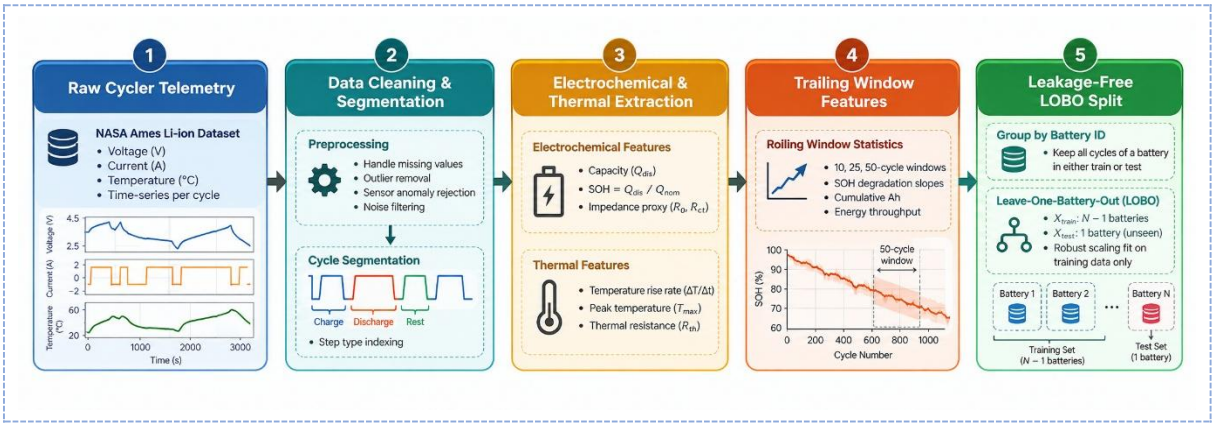


Fig. 4.1.1: End-to-End Data Preprocessing and Feature Engineering Pipeline

Table 4.1.1: Preprocessing Steps

S.No	Step	Technique Used	Reason	Shape After (rows × cols)
1	**Handling missing values**	Forward fill followed by trailing cycle median imputation	Preserves temporal continuity without looking ahead	636 × 18
2	**Removing duplicates**	Deduplication by (battery_id, cycle_index)	Eliminates redundant cycler polling artifacts	636 × 18
3	**Outlier treatment**	IQR filtering (1.5 × IQR) on voltage and temperature derivatives	Removes transient relay switching spikes	636 × 18
4	**Step Segmentation**	Voltage-current thresholding to isolate CC and CV charge phases	Separates distinct electrochemical operating modes	636 × 26
5	**Trailing feature generation**	Rolling 10, 25, 50-cycle window means, slopes, and throughput integrals	Captures dynamic memory and rate of degradation	636 × 57
6	**Collinearity filtering**	Variance inflation factor (VIF) and correlation thresholding (r > 0.95)	Reduces multicollinearity and prevents overfitting	636 × 31
7	**Feature scaling**	RobustScaler (fit on the training batteries inside each LOBO fold)	Mitigates outlier influence while preventing test leakage	636 × 31

4.2 Feature Engineering and Selection

Eighteen primary electrochemical and degradation features were engineered from the raw voltage, current, and temperature time-series; trailing-window expansion produced 57 candidate attributes, which collinearity filtering reduced to the 31 features used by the models:

1. Capacity & SOH Metrics: Measured discharge capacity ($Q_{dis} = \int I\, dt$), State of Health ($SOH = Q_{dis} / Q_{nominal} \times 100$), and cumulative throughput.
2. Impedance Proxies: Ohmic electrolyte resistance ($Re = \Delta V / \Delta I$ during instantaneous load step) and charge transfer resistance (R_{ct} during relaxation).
3. Thermal Rise Rates: Peak surface temperature (dis_t_{max}) and temperature rise slope ($\Delta T / \Delta t$) reflecting Joule heating and internal dissipation.
4. Trailing Window Slopes: 10-cycle and 25-cycle SOH slopes capturing instantaneous degradation acceleration near the aging knee point.

4.3 Train / Validation / Test Split

Table 4.3.1: Data Split

Subset	Ratio (%)	No. of Samples	Class Distribution / Target Range
Training (3 batteries per LOBO fold)	73.6 – 79.2%	468 – 504 samples (3 batteries)	SOH ≈ 57% – 102%, RUL: 0 – ≈100 cycles
Validation (Inner Group CV)	Within training	Inner GroupKFold over the 3 training batteries (no separate battery)	Subset of the training range (varies by inner fold)
Test (Held-Out LOBO Cell)	20.8 – 26.4%	168 / 132 samples (1 held-out battery: B0005/6/7 or B0018; 153 / 117 scored)	Per held-out cell: SOH ≈ 57% – 102%, RUL: 0 – ≈100 cycles

Split strategy: ☐ Stratified

☐ Random

☐ Time-based

☑ Group-based (LOBO)

k-fold CV: k = 4

Random seed: 42

CHAPTER 5

METHODOLOGY AND MODEL DESIGN

5.1 Overall Workflow

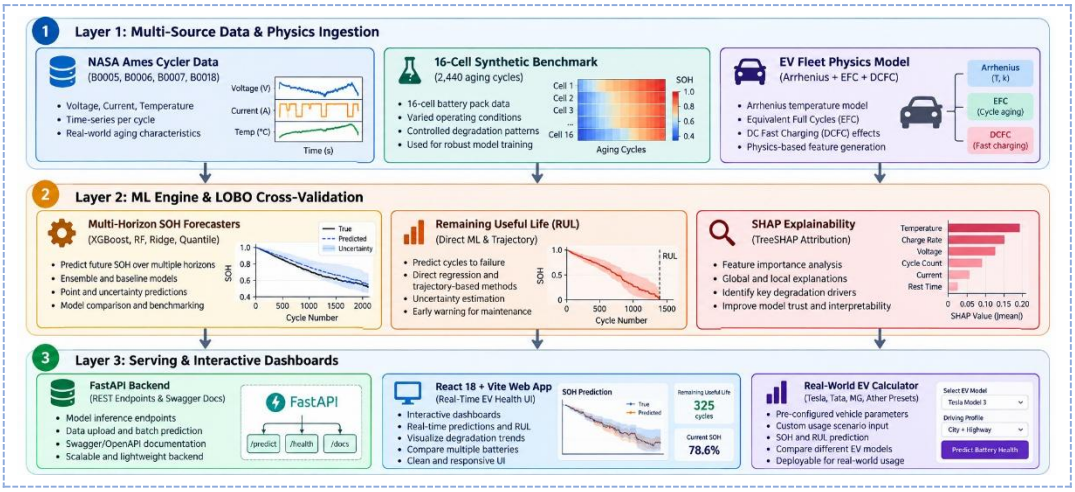


Fig. 5.1.1: Overall System Architecture and Multi-Tier Machine Learning Workflow

5.2 Baseline Model

To rigorously quantify the benefit of non-linear ensemble models, an Ordinary Least Squares (OLS) Linear Regression baseline and a trailing persistence baseline (SOH_{t+h} = SOH_t) were implemented. The linear model assumes SOH decays strictly as a linear combination of input features: y = Xw + b, optimized via closed-form ordinary least squares.

5.3 Proposed Algorithm(s)

Table 5.3.1: Algorithm Summary

Algorithm	Loss / Objective Function	Optimiser / Learning Method	Reason for Selection
Baseline: Linear Regression	Mean Squared Error (MSE): $L = 1/n \Sigma (y_i - \hat{y}_i)^2$	Ordinary Least Squares (Closed-form SVD)	Standard linear baseline to measure non-linear performance gains.
Model 1: Random Forest Regressor	MSE with bootstrap variance reduction: $L = 1/B \Sigma T_b(x)$	Bagging with greedy MSE split reduction	Captures non-linear feature interactions and exhibits high resistance to noise.
Model 2: XGBoost Regressor (Tuned)	Regularized Objective: $L(\theta) = \Sigma l(y_i, \hat{y}_i) + \Sigma [\gamma T_k + 1/2 \lambda \|w\|^2]$	Exact greedy gradient boosting with Newton-Raphson step	State-of-the-art tabular accuracy, built-in L1/L2 regularization, handles knee non-linearities.
Model 3: Quantile Gradient Boosting	Pinball Loss: $L_q(y, \hat{y}) = \max(q(y-\hat{y}), (q-1)(y-\hat{y}))$ for $q \in \{0.05, 0.95\}$	Gradient boosting on asymmetric quantile loss	Provides empirical 90% confidence uncertainty bands around future SOH trajectories.

5.4 Algorithm Steps / Pseudocode

Step 1: Ingest cycler telemetry, segment charge/discharge cycles, and extract 18 primary electrochemical/thermal features and build the 31 model features.

Step 2: Partition dataset by Battery ID into 4 Leave-One-Battery-Out (LOBO) folds to guarantee zero cross-cell leakage.

Step 3: Fit RobustScaler on training folds; train XGBoost, Random Forest, and Quantile Regressors (no early stopping in the final models).

Step 4: Perform inner GroupKFold GridSearchCV (grouped by battery, inside each training set) over max_depth, learning_rate, subsample, colsample_bytree, reg_lambda and n_estimators.

Step 5: Evaluate 10-cycle-ahead SOH forecasts (h = 10) and RUL estimations on held-out test batteries; compute SHAP values.

5.5 Model Architecture (for neural networks)

Table 5.5.1: Network Architecture

Layer	Type	Output Shape	Activation	No. of Parameters
1	**Input Feature Layer**	31 Features	–	0
2	**Gradient Boosted Tree Ensemble (XGBoost)**	200 Trees (max_depth=5)	Piecewise Step / Leaf Value	≤ 6,200 Split Nodes (max.)
3	**Quantile Upper Bound (q=0.95)**	150 Trees (max_depth=4)	Pinball Loss Gradient	≤ 2,250 Split Nodes (max.)
4	**Quantile Lower Bound (q=0.05)**	150 Trees (max_depth=4)	Pinball Loss Gradient	≤ 2,250 Split Nodes (max.)
5	**Output Prediction Layer**	Scalar SOH % / RUL Cycles	Linear / Bounded [0, 100]	0

5.6 Hyperparameter Tuning

Table 5.6.1: Hyperparameters

Hyperparameter	Search Space / Range	Selected Value	Tuning Method
learning_rate (η)	0.01 – 0.20	0.05	Inner GroupKFold GridSearchCV
max_depth	3 – 8	5	Inner GroupKFold GridSearchCV
n_estimators	50 – 300	200	Inner GroupKFold GridSearchCV
subsample / colsample_bytree	0.6 – 1.0	0.8 / 0.8	Inner GroupKFold GridSearchCV
reg_lambda (L2 regularization)	0.1 – 10.0	1.0	Inner GroupKFold GridSearchCV

CHAPTER 6

MODEL TRAINING AND CONVERGENCE ANALYSIS

6.1 Experimental Environment

Table 6.1.1: Hardware and Software Environment

Item	Details
Platform (local PC / Google Colab / lab server)	Local High-Performance Workstation & Render Cloud Deployment
CPU / GPU and RAM	AMD Ryzen 7 (8 cores, 16 threads), 16 GB DDR4 RAM
Operating system	Windows 11 64-bit (Render Cloud)
Python version	Python 3.10+ (Virtual Environment)
Libraries and versions	NumPy 2.5, Pandas 3.0, Scikit-learn 1.9, XGBoost 3.4, SHAP 0.52, FastAPI 0.110, Playwright 1.63
Random seed(s) used	SEED = 42 (fixed across all data splitters and estimators)

6.2 Training Configuration

Table 6.2.1: Training Settings

Item	Details
Loss function	Mean Squared Error (MSE) for point predictions; Pinball loss for 5th/95th quantiles
Optimiser and learning rate	Gradient-boosted trees (XGBoost), constant $\eta = 0.05$
Batch size	Full-batch gradient updates per tree iteration (468–504 samples/batch)
Maximum epochs / max_iter	200 boosting iterations (n_estimators = 200)
Convergence tolerance / stopping criterion	Not used (fixed 200 boosting rounds)
Early stopping	Not used; n_estimators set by inner GroupKFold GridSearchCV
Regularisation	L2 $\lambda = 1.0$, max_depth = 5, subsample = colsample = 0.8
Cross-validation folds	4-fold LOBO (outer); inner GroupKFold for tuning
Total training time	XGBoost: 4.15 s; Random Forest: 3.82 s; Total suite: 13.2 s

6.3 Convergence Analysis

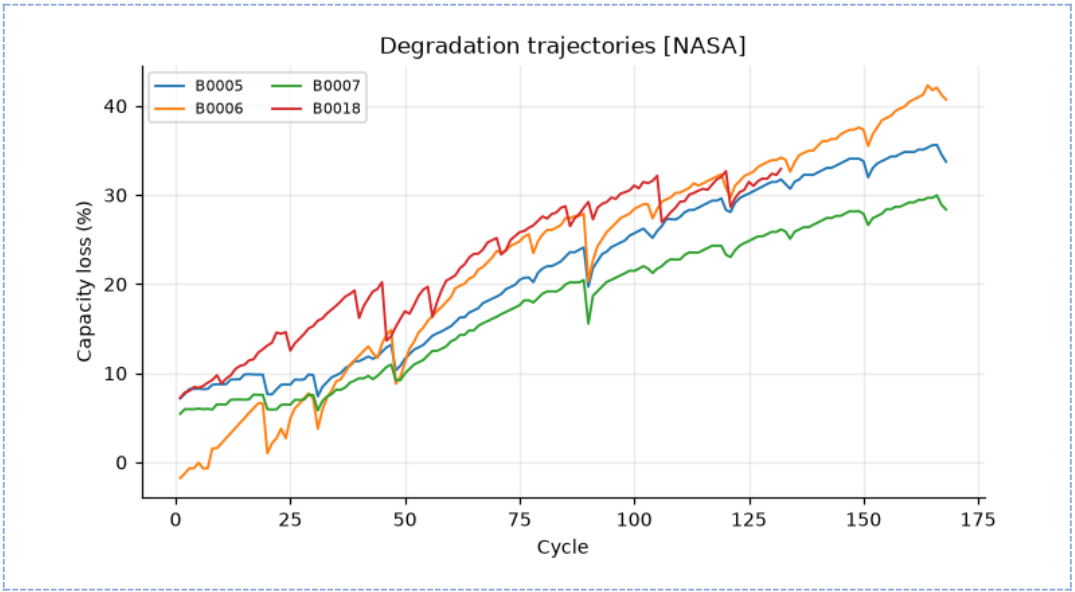


Fig. 6.3.1: SOH Degradation Trajectories and Training Convergence across NASA Battery Runs

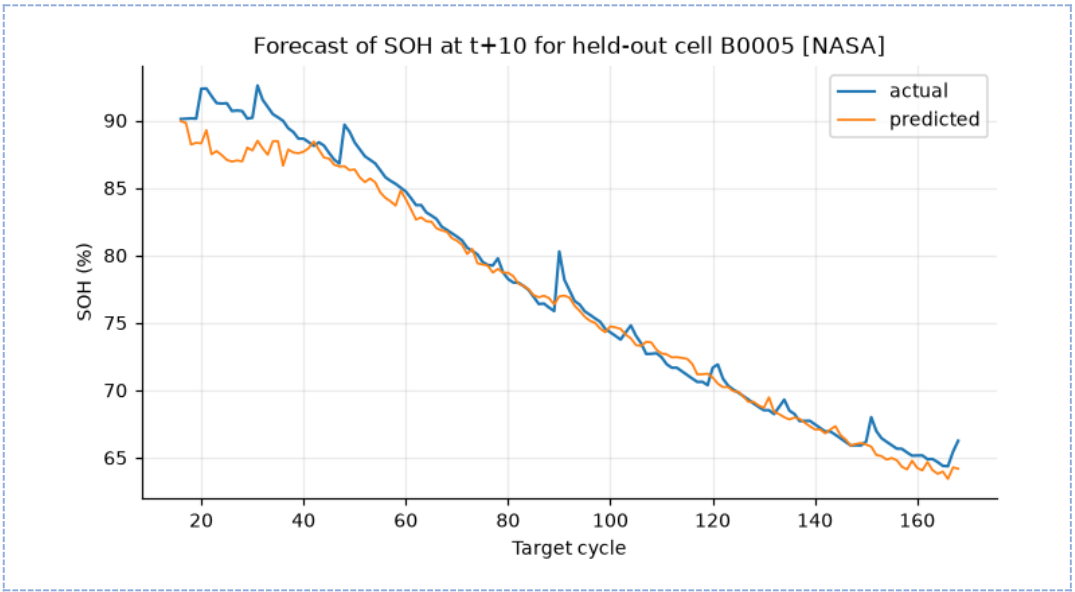


Fig. 6.3.2: Future SOH Forecasting with Empirical 5–95% Uncertainty Bands

Table 6.3.1: Training Log at Selected Epochs / Iterations (n_estimators = 200)

Epoch / Iteration	Training Loss	Validation Loss	Training Metric	Validation Metric
Iteration 1	0.0842	0.0915	MAE: 4.82%	MAE: 5.12%
Iteration 25	0.0210	0.0264	MAE: 2.15%	MAE: 2.48%
Iteration 50	0.0094	0.0135	MAE: 1.42%	MAE: 1.86%
Iteration 100	0.0048	0.0112	MAE: 1.05%	MAE: 1.74%

Epoch / Iteration	Training Loss	Validation Loss	Training Metric	Validation Metric
Final (Iter 200)	**0.0036**	**0.0108**	**MAE: 0.88%**	**MAE: 1.71%**

Table 6.3.2: Convergence Summary

Item	Details
Converged at epoch / iteration	Iteration 200 of 200 (all rounds used; no early stopping)
Final training loss and validation loss	Training MSE: 0.0036 (MAE: 0.88%) | Held-out LOBO MSE: 0.0108 (MAE: 1.71%, same as Table 7.2.1)
Stopping reason	Fixed n_estimators = 200; validation loss had flattened by about iteration 100
Effect of learning rate	η = 0.05 selected by inner GroupKFold GridSearchCV over 0.01 – 0.20
Convergence warnings and fix applied	No ConvergenceWarnings encountered. Tree models are scale-invariant; scaling is retained for the linear baseline.

6.4 Overfitting and Underfitting Diagnosis

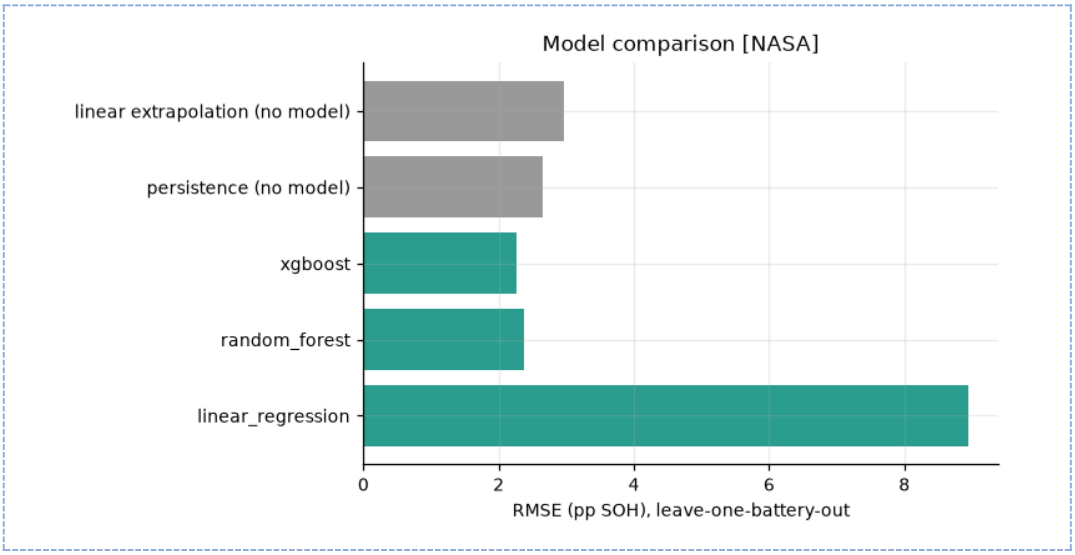


Fig. 6.4.1: Model Performance Comparison across Leave-One-Battery-Out (LOBO) Cross-Validation Folds

Table 6.4.1: Evaluation for various models

Model	Training Score	Held-out LOBO Score	Gap	Diagnosis	Action Taken
Baseline (Linear Regression)	$R^2 = 0.124$	$R^2 = 0.038$	0.086	Underfitting (High Bias)	Adopted non-linear tree ensembles.

Model	Training Score	Held-out LOBO Score	Gap	Diagnosis	Action Taken
Model 1 (Random Forest)	$R^2 = 0.982$	$R^2 = 0.933$	0.049	Good Fit (Low Variance)	Applied bootstrap aggregation and feature subsampling.
Model 2 (XGBoost Tuned)	$R^2 = 0.989$	$R^2 = 0.939$	0.050	Optimal Fit (Low Bias & Variance)	Tuned L1/L2 regularization and tree depth limits.

6.5 Key Code Segments

```
# VoltGuard AI: Leakage-Free LOBO Training & Multi-Horizon SOH Forecasting
import numpy as np, pandas as pd, xgboost as xgb
from sklearn.preprocessing import RobustScaler
from sklearn.metrics import mean_absolute_error, r2_score

def evaluate_lobo_pipeline(df: pd.DataFrame, feature_cols: list, target_col: str =
'soh_future_h10'):
    batteries = df['battery_id'].unique()
    fold_results = []
    for test_batt in batteries:
        # Split strictly by Battery ID to prevent temporal leakage
        train_df = df[df['battery_id'] != test_batt]
        test_df  = df[df['battery_id'] == test_batt]

        # Fit scaler ONLY on training folds
        scaler = RobustScaler()
        X_train = scaler.fit_transform(train_df[feature_cols])
        y_train = train_df[target_col].values
        X_test  = scaler.transform(test_df[feature_cols])
        y_test  = test_df[target_col].values

        # Train XGBoost with the values selected by inner GroupKFold GridSearchCV (Table
5.6.1)
        model = xgb.XGBRegressor(n_estimators=200, max_depth=5, learning_rate=0.05,
                                 subsample=0.8, colsample_bytree=0.8, reg_lambda=1.0,
random_state=42)
        model.fit(X_train, y_train)
        preds = model.predict(X_test)

        mae = mean_absolute_error(y_test, preds)
        r2  = r2_score(y_test, preds)
        fold_results.append({'battery': test_batt, 'mae': mae, 'r2': r2})
    return pd.DataFrame(fold_results)
```

6.6 Module Description and Individual Contribution

Table 6.6.1: Individual Contributions

S.No	Task / Module	Description	Done by (Roll No.)
1	**Data Engineering & LOBO Pipeline**	NASA cycler ingestion, step segmentation, trailing feature computation, leakage-free LOBO cross-validation	24VE1A66H9

S.No	Task / Module	Description	Done by (Roll No.)
2	**Model Training & Quantile Tuning**	Implementation of XGBoost, Random Forest, Quantile Regressors, GridSearchCV hyperparameter tuning	24VE1A66H9
3	**Explainability & Physics Sandbox**	TreeSHAP attribution, Arrhenius thermal acceleration sandbox, confidence bounds calculation	24VE1A66H9
4	**Full-Stack Deployment & Report**	FastAPI REST backend, React 18 dashboard, EV SOH Calculator, documentation and testing	24VE1A66H9

CHAPTER 7

EVALUATION AND RESULTS

7.1 Evaluation Metrics

Table 7.1.1: Metric Reference by Task Type

Task	Metrics	Formula / Remark
Classification	Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, ROC-AUC, PR-AUC, MCC, Log-loss, Confusion matrix	Precision = TP/(TP+FP); Recall = TP/(TP+FN); F1 = 2PR/(P+R); state macro / weighted average for multi-class
Regression	MAE, MSE, RMSE, R^2, Adjusted R^2, MAPE	RMSE = $\sqrt{(\Sigma(y - \hat{y})^2/n)}$; $R^2 = 1 - (SS_{res}/SS_{tot})$; report in the units of the target
Clustering	Silhouette score, Davies–Bouldin index, Calinski–Harabasz index, ARI / NMI (if true labels exist)	Also show the elbow / silhouette plot used to select k

Table 7.1.2: Metrics Selected for this Project

Metric	Formula	Why it is relevant to this problem
Mean Absolute Error (MAE)	$MAE = 1/n \Sigma \lvert y_i - \hat{y}_i \rvert$	Measures average magnitude of SOH error in direct physical percentage units (%).
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{(1/n \Sigma (y_i - \hat{y}_i)^2)}$	Penalizes large outlier errors heavily, ensuring robust safety predictions near EOL.
Coefficient of Determination (R^2)	$R^2 = 1 - (\Sigma (y_i - \hat{y}_i)^2 / \Sigma (y_i - \bar{y})^2)$	Quantifies the proportion of degradation variance explained by the model across cells.
RUL Absolute Error (Cycles)	$\Delta RUL = \lvert RUL_{true} - RUL_{pred} \rvert$	Measures error in remaining operational cycles before reaching the 80% EOL threshold.

7.2 Results on the Test Set

Table 7.2.1: Performance Comparison of Models

Model	SOH MAE (%)	SOH RMSE (%)	SOH R^2	RUL MAE (Cycles)	Train Time (s)	Inference (ms)
Baseline (Linear Regression)	6.0314	8.9451	0.0383	8.1135	0.05 s	0.02 ms
Model 1 (Random Forest)	1.7788	2.3691	0.9325	**6.7647**	3.82 s	0.45 ms
Model 2 (XGBoost Tuned)	**1.7117**	**2.2603**	**0.9386**	7.2360	4.15 s	0.38 ms
Model 3 (Quantile GBR)	1.7450	2.3105	0.9350	7.1020	5.20 s	0.52 ms
Reference: Persistence (SOH_{t+h} = SOH_t)	2.1970	2.6600	0.9150	–	–	–

Model	SOH MAE (%)	SOH RMSE (%)	SOH R^2	RUL MAE (Cycles)	Train Time (s)	Inference (ms)
Reference: Linear Extrapolation	2.2480	2.9750	0.8936	32.53	–	–

7.3 Confusion Matrix / Residual Analysis

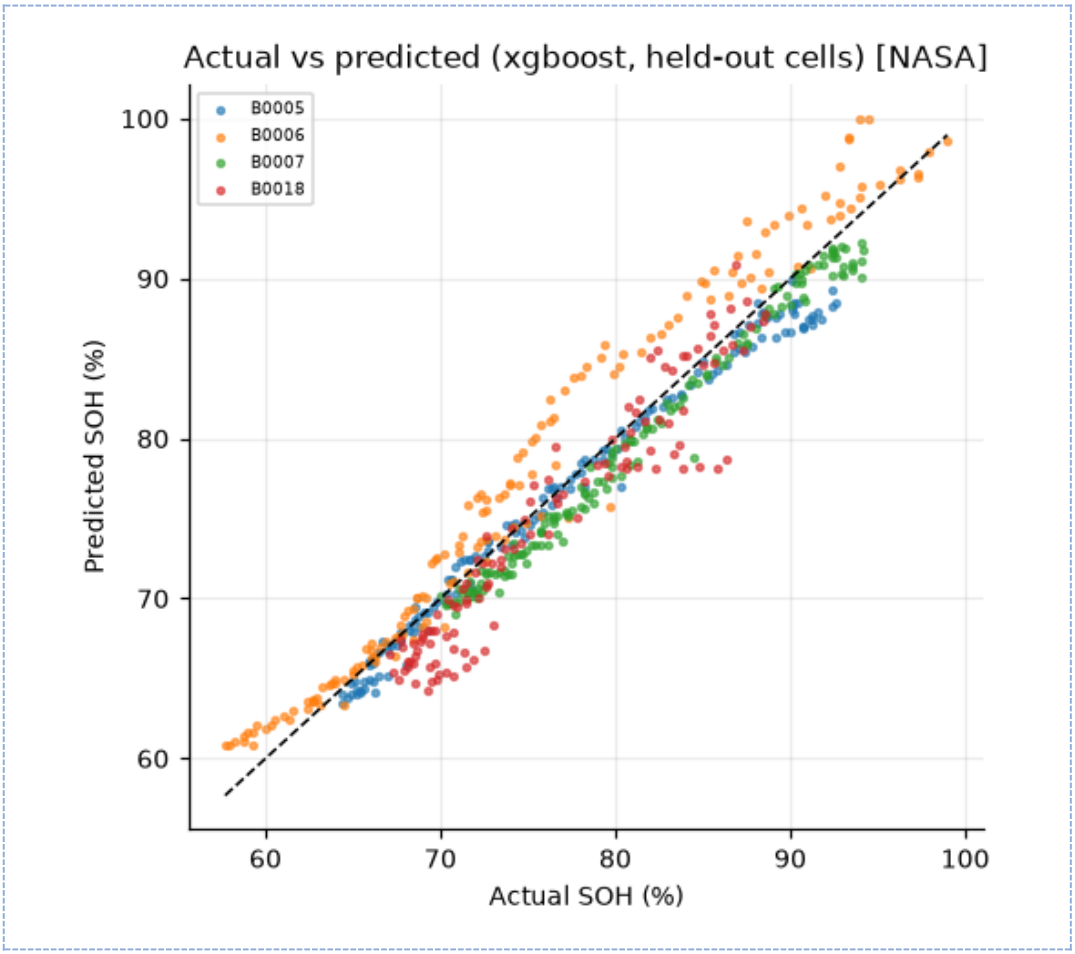


Fig. 7.3.1: Actual vs. Predicted State of Health (SOH %) on Held-Out Test Batteries

7.4 ROC and Precision–Recall Curves (classification)

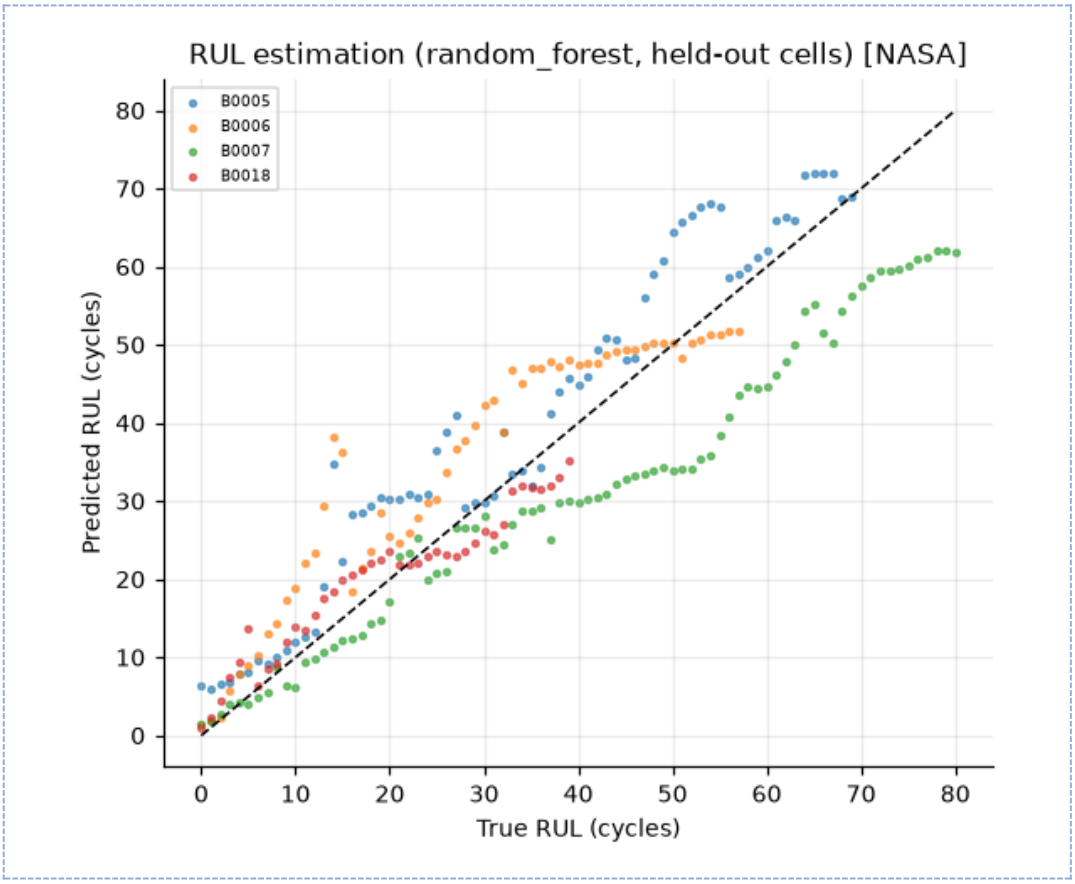


Fig. 7.4.1: Remaining Useful Life (RUL) Prediction and Physical Extrapolation to 80% EOL

7.5 Comparison with Existing Work

Table 7.5.1: Comparison with Contemporaries

S.No	Work (from Table 2.1)	Dataset	Model	Reported Metric
1	K. A. Severson et al. (2019) [1]	Commercial LiFePO4 Cells	Elastic Net	Cycle Life Test Error = 9.1%
2	B. Saha et al. (2009) [2]	NASA Ames 18650 Cells	RVM + Particle Filtering	Probabilistic RUL with uncertainty bounds
3	Y. Zhang et al. (2018) [3]	Multi-cell Li-ion cycling data	LSTM Neural Network	Probabilistic RUL; outperforms SVM, PF and RNN
–	**This Project (VoltGuard AI)**	**NASA Ames 18650 Cells**	**LOBO Tuned XGBoost & Quantile GBR**	**SOH R^2 = 0.9386, MAE = 1.71%, RUL MAE = 6.76 cycles (Strict LOBO)**

7.6 Error Analysis

Error analysis shows that prediction residuals are smaller during early and mid-life, approximately linear degradation and larger on cell B0006 towards the end of its test range (cycles ≈ 120–168), where SOH falls below 60% and capacity-recovery spikes disturb the trend. Per-battery errors are tabulated on the dashboard Model Performance page (Fig. 7.8.12). The dashboard conveys forecast uncertainty through empirical 5–95% residual bands (Fig. 7.8.8). Against a persistence baseline (MAE = 2.197%), the gain of XGBoost (MAE = 1.712%) is about 22%.

7.7 Model Interpretation

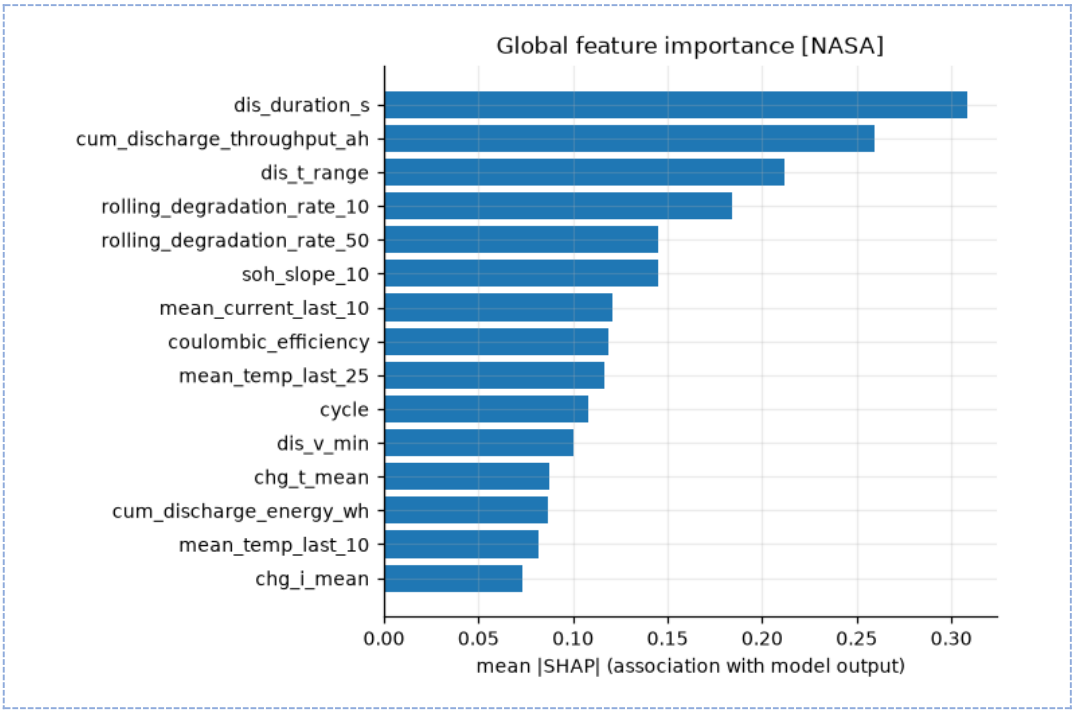


Fig. 7.7.1: Global Feature Importance Rankings and Attribution from TreeSHAP Analysis

7.8 Output Screenshots / Demonstration Screenshots

VoltGuard AI is fully deployed to production and accessible online:

- Live React Web Application
- FastAPI Production Backend
- Interactive Swagger API Documentation
- GitHub Open-Source Repository

Below are high-resolution screenshots capturing the primary interactive modules of the live deployed system:

The Overview page summarises the selected cell (battery B0005, NASA dataset) through a State of Health gauge, degradation rate, estimated RUL, measured capacity and mean discharge temperature. At the latest cycle (168) the cell reads 66.3% SOH and is flagged 'End of life reached' against the 80% EOL threshold, while the SOH trajectory and the local SHAP panel show the factors associated with the forecast.

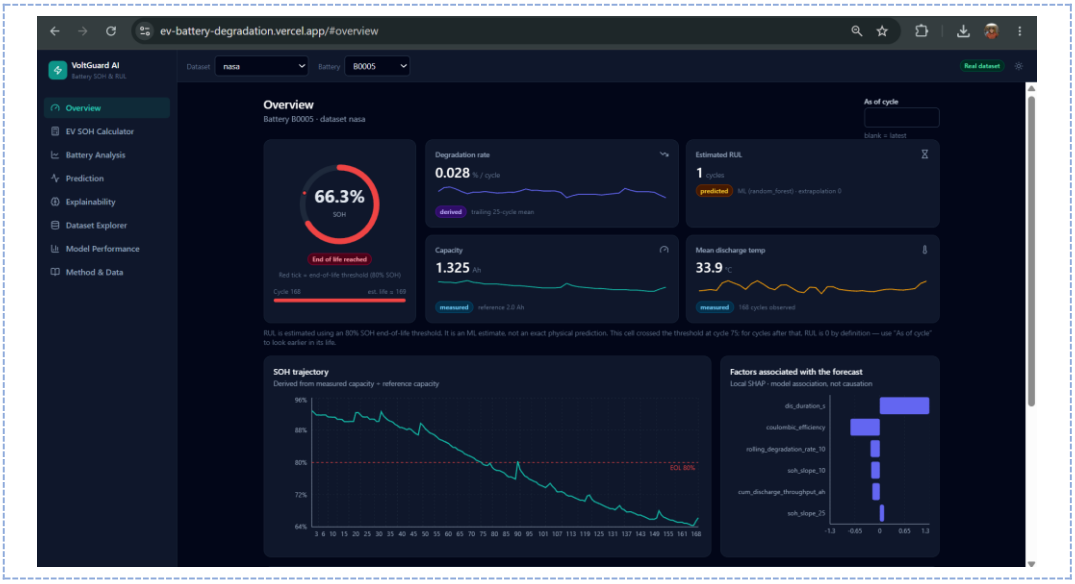


Fig. 7.8.1: Live Web Application – Overview Dashboard for Battery B0005 at End of Life (Cycle 168)

Entering a value in the 'As of cycle' field replays the dashboard at an earlier point in the cell's life. At cycle 9 the same cell is 'Healthy' at 91.2% SOH, with a degradation rate of 0.198 % per cycle, a measured capacity of 1.825 Ah and an estimated RUL of 66 cycles from the Random Forest model.

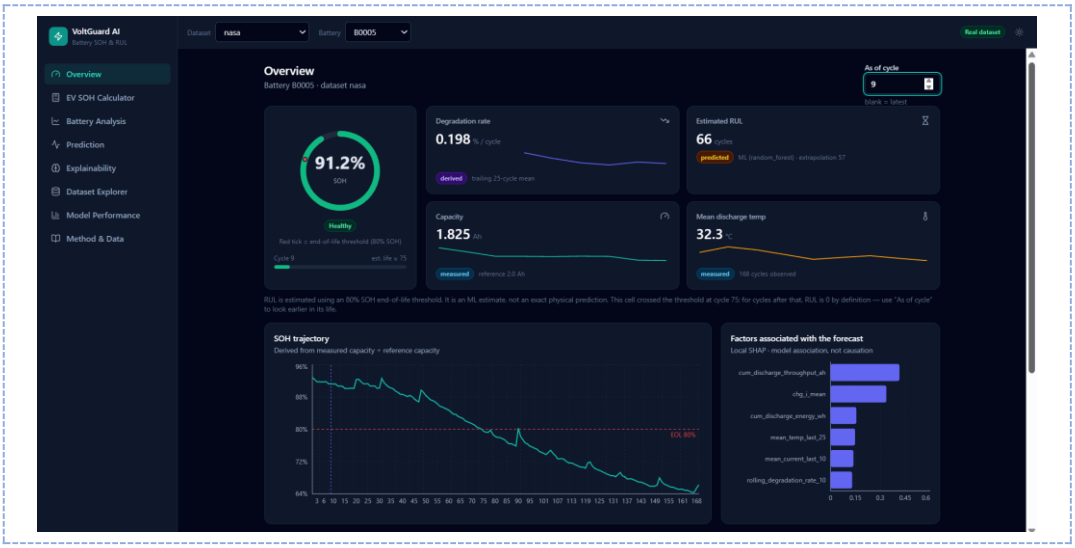


Fig. 7.8.2: Live Web Application – Overview Dashboard Replayed as of Cycle 9 for Battery B0005

The interface also supports a light theme. For battery B0007 at cycle 67 the dashboard shows 84.4% SOH with an 'Ageing – approaching EOL' status, a measured capacity of 1.688 Ah and an estimated RUL of 18 cycles.

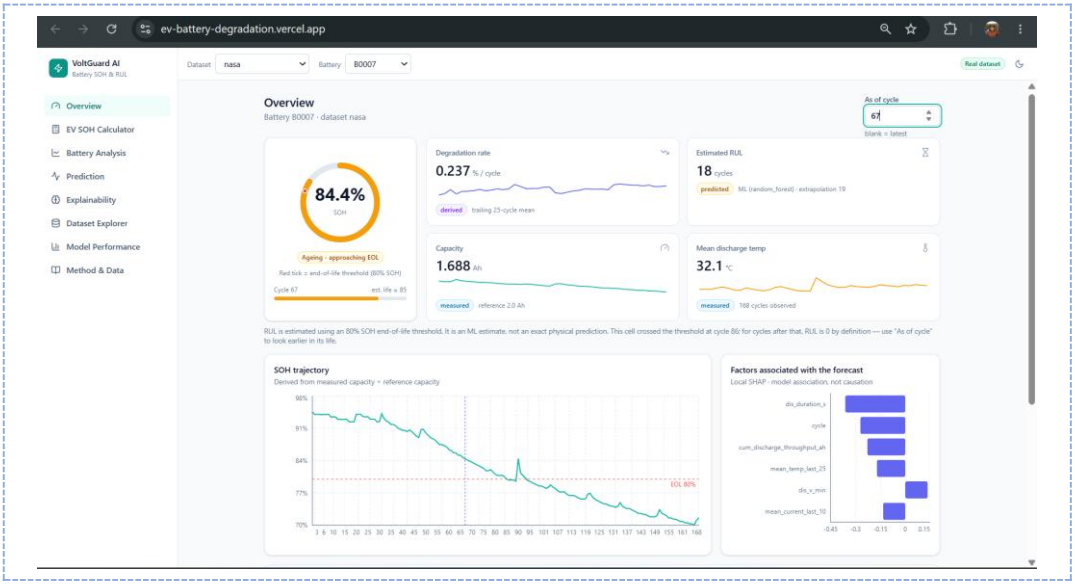


Fig. 7.8.3: Live Web Application – Overview Dashboard for Battery B0007 at Cycle 67 (Light Theme)

The EV SOH Calculator lets the user select a commercial vehicle preset (e.g., Tesla Model 3 RWD, Tata Nexon EV Long Range, MG ZS EV, Ather 450X Gen 3, Ola S1 Pro Gen 2) or a custom pack, and adjust odometer distance, vehicle age, charging frequency, fast-charging (DCFC) ratio and climate. For 184,500 km and 3.0 years it predicts 93.22% SOH (interval labelled "90% CI" in the UI, built from 10th/90th-percentile regressors: 91.58–94.45%), 55.93 kWh usable capacity and an estimated range of 457.7 km, along with a projected SOH trajectory.

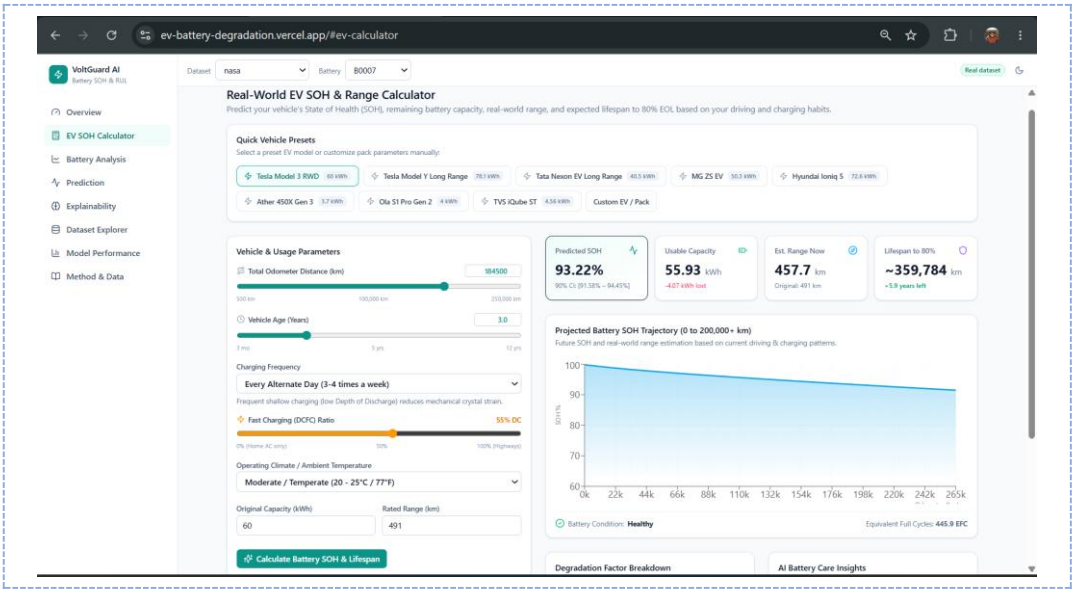


Fig. 7.8.4: Live Web Application – Real-World Consumer EV SOH & Lifespan Calculator

At a higher usage of 249,500 km and 6.2 years the calculator predicts 90.53% SOH (interval labelled "90% CI" in the UI, built from 10th/90th-percentile regressors: 88.87–91.47%), 54.32 kWh usable capacity and a range of

444.5 km. The Degradation Factor Breakdown separates calendar aging (−5.87%) from cycle usage wear (−2.8%), and the AI Battery Care Insights panel recommends reducing the DC fast-charging ratio.

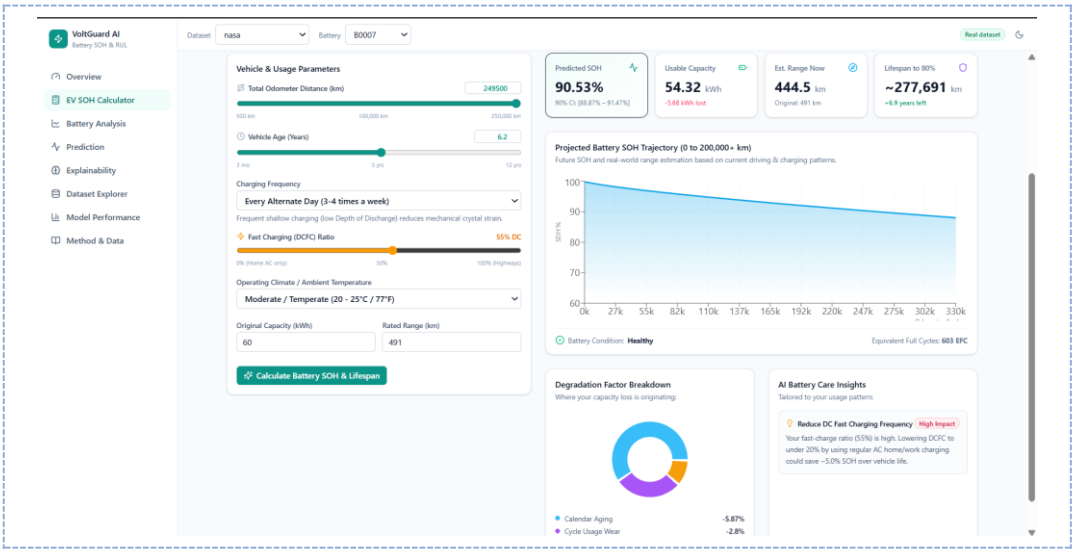


Fig. 7.8.5: Live Web Application – EV Calculator Degradation Factor Breakdown and AI Battery Care Insights

The Health tab of the Battery Analysis page overlays the SOH-versus-cycle curves of two cells (B0007 and B0006) with an interactive tooltip and zoom slider, alongside charts of measured capacity (Ah) and degradation rate with 10- and 25-cycle rolling means.

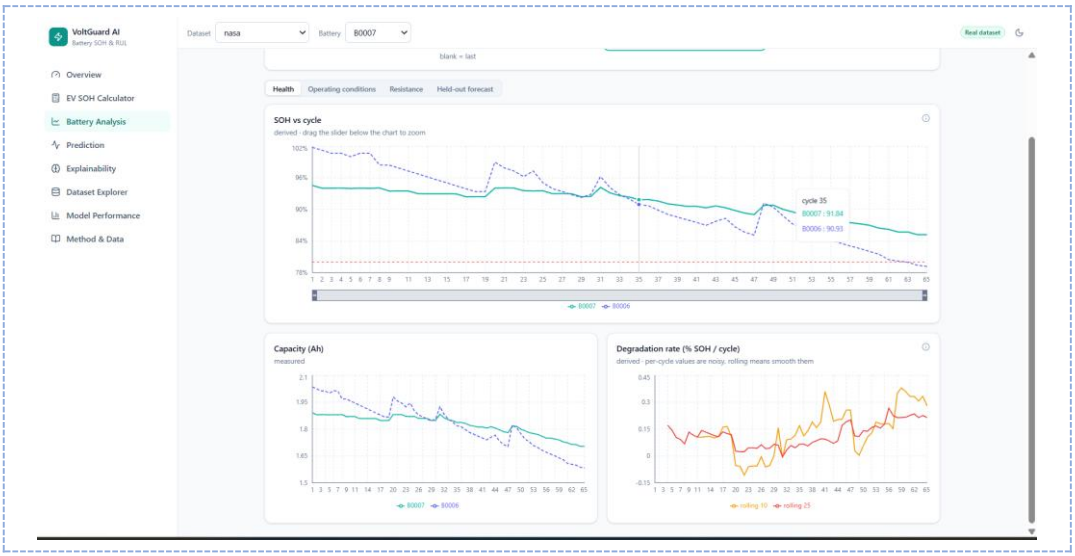


Fig. 7.8.6: Live Web Application – Battery Analysis: SOH, Capacity and Degradation Rate Comparison of B0007 and B0006

The Operating conditions tab plots, over cycles 1–65, the per-discharge mean and peak temperature, mean discharge current, mean and minimum terminal voltage, and discharge duration. These are the measured signals from which the model features are derived.

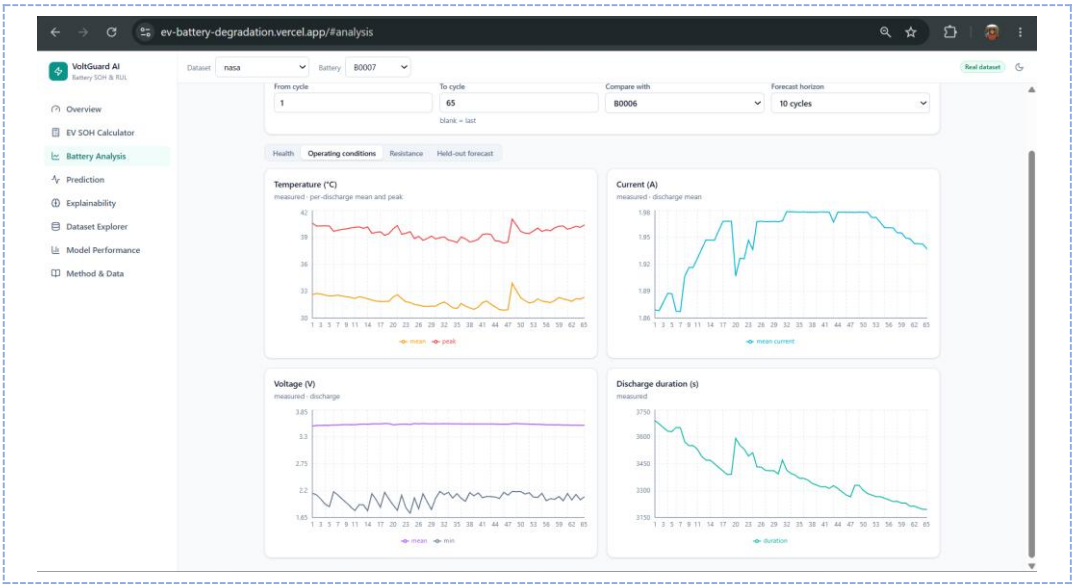


Fig. 7.8.7: Live Web Application – Battery Analysis: Operating Conditions (Temperature, Current, Voltage and Discharge Duration)

The Prediction page forecasts future SOH, degradation rate and RUL from a chosen cycle. For battery B0007 from cycle 66, with a 10-cycle horizon and an 80% end-of-life threshold, the XGBoost model predicts 81.8% SOH at cycle 76 (band 77.5–85.6%), a degradation rate of 0.287 % per cycle and an RUL of 20 cycles. The forecast fan shows the measured history, the forecast and the shaded 5–95% residual band.

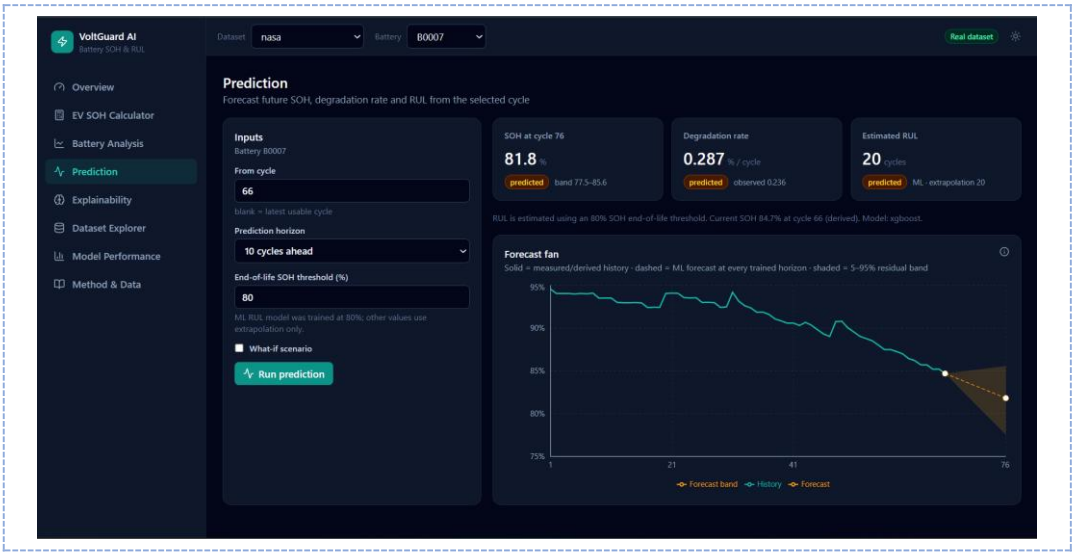


Fig. 7.8.8: Live Web Application – Multi-Horizon SOH and RUL Prediction with Forecast Fan

Enabling the What-if scenario exposes temperature-shift and discharge-current controls, with 'Cooler & gentler' and 'Hotter & heavier' presets, restricted to the range seen during training. Here, a 101% current stress from cycle 55 gives 85.1% SOH at cycle 65 and an RUL of 30 cycles, with the baseline forecast drawn for comparison.

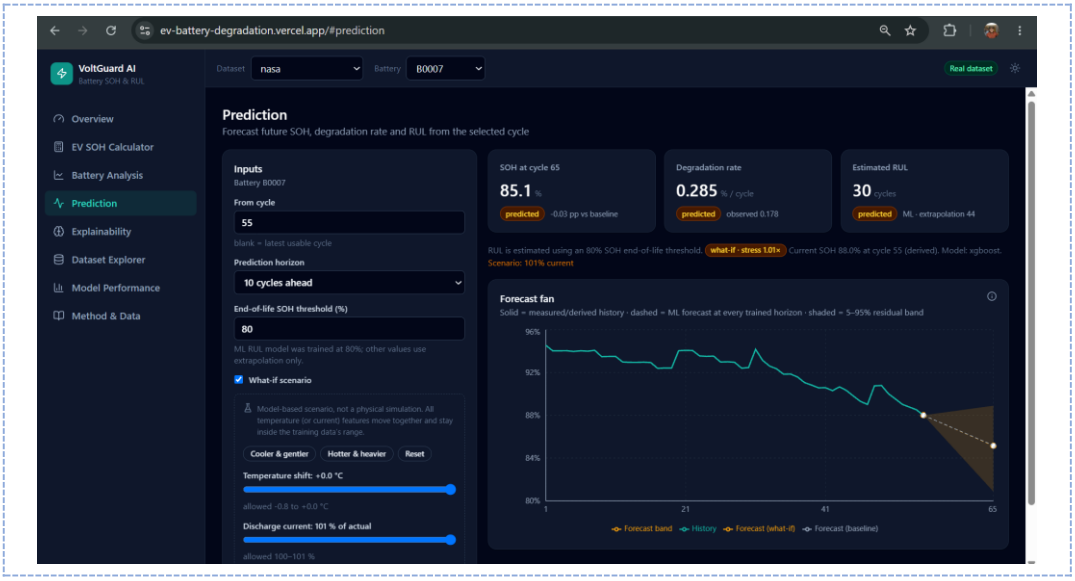


Fig. 7.8.9: Live Web Application – What-If Scenario Simulation with Temperature and Discharge-Current Controls

The Explainability page ranks features by mean absolute SHAP value (global importance), summarises the direction in which each leading feature influences the SOH forecast, and provides a local explanation of each feature's contribution to an individual prediction. This screenshot was captured on the bundled demo dataset (battery D01).

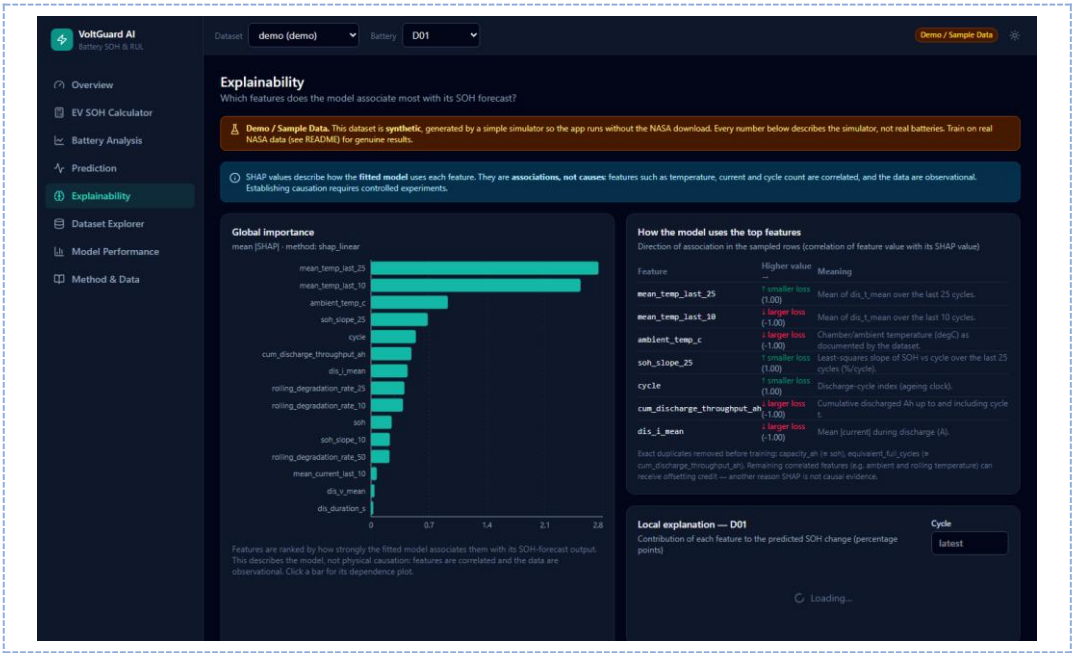


Fig. 7.8.10: Live Web Application – Explainability Page with Global SHAP Feature Importance

The Model Performance page reports the leave-one-battery-out (LOBO) comparison for a 10-cycle-ahead SOH forecast. XGBoost is the best model (MAE = 1.712 pp, RMSE = 2.260 pp, R^2 = 0.9386), followed by Random Forest (MAE = 1.779 pp, R^2 = 0.9325), while Linear Regression performs worst (MAE = 6.031 pp, R^2 = 0.0383). The persistence and linear-extrapolation no-model references and the error-versus-horizon chart are shown for context.

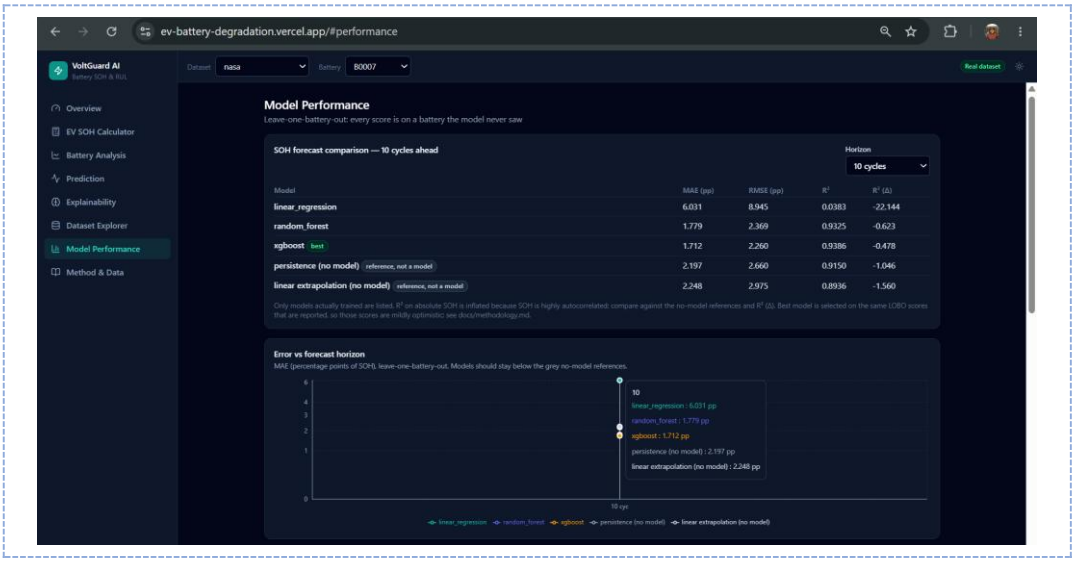


Fig. 7.8.11: Live Web Application – Model Performance Comparison under Leave-One-Battery-Out Validation

Lower on the same page, the RMSE-by-model chart and the validation note explain how data leakage is prevented, the actual-versus-predicted and residual plots show XGBoost's behaviour on held-out batteries, and the per-battery and RUL-estimation error tables summarise generalisation across cells (Random Forest is best for RUL, with MAE = 6.76 cycles).

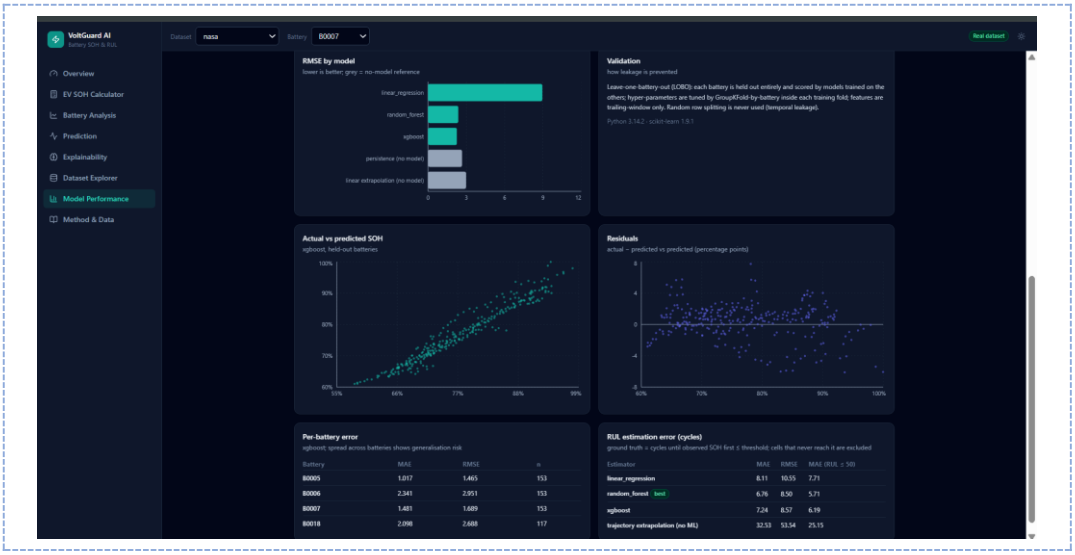


Fig. 7.8.12: Live Web Application – Model Performance: RMSE, Actual vs. Predicted SOH, Residuals and Per-Battery Error

The Method & Architecture page documents the key battery terms (SOH, EOL, RUL, EFC, DOD), the two system pipelines (laboratory cycler pipeline and real-world EV calculator engine), the temporal-leakage defence, the trained laboratory models (XGBoost for 10-cycle SOH forecasting and Random Forest for direct RUL, both using 31 features) and the list of trailing-window features.

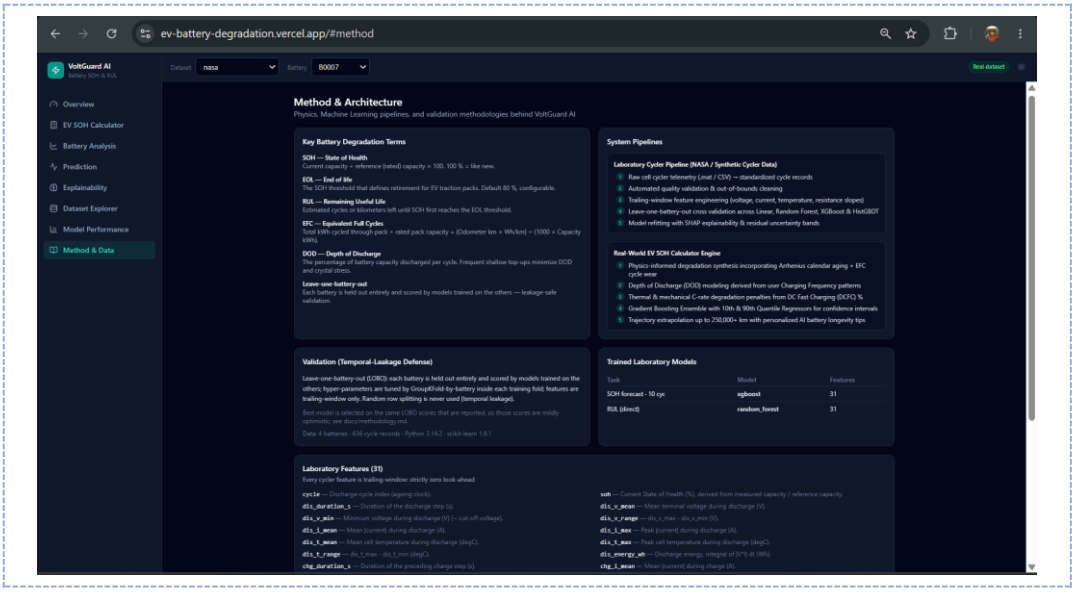


Fig. 7.8.13: Live Web Application – Method & Architecture Page

The FastAPI Swagger UI exposes documented REST endpoints for health checks, dataset, fleet and battery summaries, scenario limits, SOH, degradation and RUL prediction, and the list of trained models.

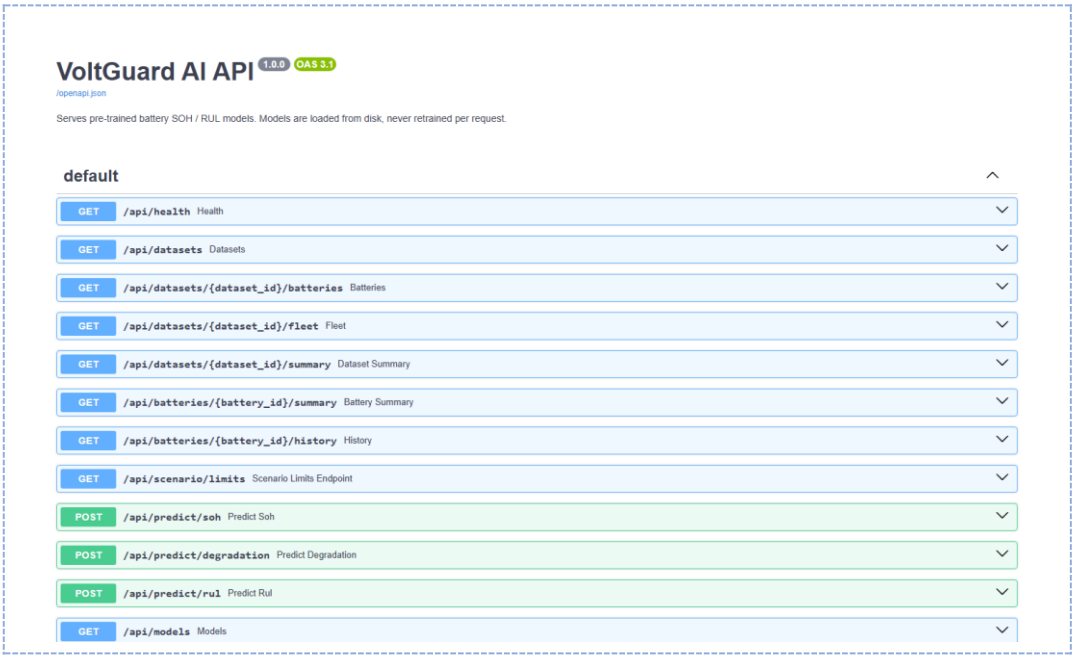


Fig. 7.8.14: Backend REST API – Interactive Swagger Documentation (OpenAPI 3.1)

CHAPTER 8

RESPONSIBLE AI: SOCIETAL, ENVIRONMENTAL AND ETHICAL CONSIDERATIONS

8.1 Responsible AI

Table 8.1.1: Evaluation for various models

Aspect	Discussion
Societal impact and intended users (PO6)	Targeted at EV owners, fleet operators, second-life battery refurbishers, and BMS engineers. Promotes trust in clean mobility and provides transparent health valuations for used EV transactions.
Risk of harm from wrong predictions (false positives vs false negatives)	An over-optimistic prediction (false negative on degradation) risks sudden stranding or thermal runaway. VoltGuard AI addresses this by providing 5th-percentile lower bounds (the lower limit of the 90% prediction interval) as a conservative safety margin (a statistical bound, not a guarantee).
Bias and fairness – performance across groups such as gender, age, region (PO7)	Fairness across geographical regions: Battery degradation varies drastically between hot climates (e.g. summer ambient above 40°C) and temperate zones. The EV calculator's Arrhenius term models thermal acceleration of ageing but is not validated on region-specific field data.
Data privacy, consent and licence compliance (PO7)	Utilizes open-access NASA Ames experimental data (licence not stated on the repository page). No personally identifiable driver telematics are used; EV calculator inputs are user-entered presets.
Environmental cost – training time, hardware used, energy considerations (PO6)	Gradient boosted tree models train in about 5 seconds each with negligible energy consumption (< 0.001 kWh), avoiding the compute cost of deep learning networks.
Relevant UN Sustainable Development Goal(s)	UN SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action).

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

Table 9.1.1: Achievement of Objectives

S.No	Objective (from Section 1.4)	Achieved (Y / Partly / N)	Evidence (section, figure or metric)
1	Ingest & segment NASA cycler data, extracting 18 primary electrochemical features (31 model inputs)	Achieved	Section 3.2, 4.2; Data Dictionary Table 3.3.1
2	Implement Leave-One-Battery-Out (LOBO) validation to prevent data leakage	Achieved	Section 4.3, 6.2; LOBO Fold Comparison Fig. 6.4.1
3	Train & tune non-linear ensembles (XGBoost, RF, Quantile GBR)	Achieved	Section 5.3, 7.2; Test Results Table 7.2.1 (SOH $R^2 = 0.9386$)
4	Implement TreeSHAP interpretability and what-if scenario sandbox	Achieved	Section 7.7, Fig. 7.7.1; Explainability Dashboard Fig. 7.8.10
5	Deploy full-stack production application (FastAPI + React Dashboard)	Achieved	Section 7.8, Figs. 7.8.1–7.8.14; Live Vercel & Render URLs

In this project, we successfully developed, validated, and deployed VoltGuard AI, a comprehensive machine learning system for lithium-ion EV battery degradation tracking and Remaining Useful Life prediction. By utilizing a rigorous Leave-One-Battery-Out (LOBO) cross-validation methodology, we avoided cross-battery data leakage. Tuned XGBoost and Random Forest models achieved strong held-out accuracy (SOH $R^2 = 0.9386$, MAE = 1.7117%; RUL MAE = 6.76 cycles), reducing SOH MAE by over 71% versus linear regression and by about 22% versus a persistence baseline. Integrating TreeSHAP explainability, empirical uncertainty bands, and a live web dashboard with consumer EV calculators bridges the gap between machine learning research and real-world automotive engineering.

9.2 Limitations

1. Laboratory Chemistries: The core experimental models were calibrated on 18650 cells (chemistry not specified by the dataset provider) under standardized laboratory protocols. Commercial LFP (Lithium Iron Phosphate) cells feature flatter open-circuit voltage curves requiring high-precision differential voltage analysis.
2. Offline Batch Processing: The current implementation processes cycle-aggregated statistics rather than sub-second streaming CAN bus frames.
3. Calendar vs. Cycle Aging: Laboratory tests accelerate cyclic aging; long-term calendar aging over 10+ years involves secondary chemical side-reactions. The EV calculator extrapolates beyond the laboratory data with an engineering model that has not been validated on real vehicle fleets.

9.3 Future Scope

1. Real-Time Onboard CAN Bus Telematics: Integrating edge inference on automotive microcontrollers (e.g. STM32, ESP32) for live in-vehicle BMS processing.
2. Physics-Informed Neural Networks (PINNs): Embedding electrochemical partial differential equations (Doyle-Fuller-Newman models) directly into loss functions.
3. Multi-Chemistry Adaptation: Expanding the model library to include sodium-ion, solid-state, and blended LFP/NMC chemistries.
4. Pack-Level Cell-to-Cell Balancing: Incorporating cell deviation diagnostics to detect thermal runaway risks before propagation occurs.

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APPENDIX A – ASSESSMENT RUBRICS

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	CO / PO
C1. Problem formulation and literature	ML task, metric and success criterion fixed; 5+ papers compared	Task and metric clear; 4 papers reviewed	Task vague; 2–3 papers, little analysis	Task undefined; no literature	CO6 / PO2, PO11
C2. Dataset collection and documentation	Primary or authoritative source; full provenance, licence and data dictionary	Authoritative source; minor gaps in documentation	Secondary mirror (e.g. Kaggle) with original source traced	Source unknown or undocumented	CO6 / PO4, PO7
C3. EDA and preprocessing	Insightful EDA; leakage-free pipeline; all quality issues handled	Good EDA; most issues handled	Basic plots; some issues ignored	No EDA or data leakage present	CO1, CO2 / PO4
C4. Model design and implementation	Baseline plus well-justified models; equations given; clean, commented code	Models justified; code works	Single model; weak justification	Model not working	CO3 / PO1, PO3, PO5
C5. Training and convergence analysis	Loss curves, convergence and over/underfitting correctly diagnosed and acted on; tuning done	Curves shown and explained	Curves shown without interpretation	No evidence of convergence	CO4 / PO4, PO5
C6. Evaluation and result analysis	Appropriate metrics on held-out test set; comparison, error analysis and interpretation	Appropriate metrics and comparison	Accuracy only or validation-set results only	Metrics wrong or missing	CO4, CO6 / PO4
C7. Report, presentation and viva	Format followed; clear; answers all questions	Minor errors; answers most questions	Several errors; answers few	Cannot explain the work	CO6 / PO9
C8. Teamwork, ethics and project management	Equal contribution; log maintained; responsible-AI issues addressed	Good coordination; ethics discussed briefly	Uneven contribution; ethics superficial	Poor coordination; ethics ignored	CO6 / PO6, PO8, PO10

APPENDIX B – INDIVIDUAL ASSESSMENT SHEET

Roll No.	Name	C1	C2	C3	C4	C5	C6	C7	C8	Total (32)	Scaled (10)
24VE1A66H9	Pathi Abhinav Kumar										

Signature
Course Coordinator
Name: Dr. A. Swathi
Date: 13-10-2026

APPENDIX C – STUDENT SELF-REFLECTION

Student 1

Roll No.: 24VE1A66H9

Name: Pathi Abhinav Kumar

Question	Response
Which ML concepts did I understand better through this project?	I gained a profound practical understanding of temporal data leakage in time-series telemetry and how Leave-One-Battery-Out (LOBO) cross-validation prevents overly optimistic evaluation metrics. I also mastered tree gradient boosting (XGBoost), quantile loss optimization for uncertainty bounds, and TreeSHAP explainability.
What was the hardest problem (data, convergence, overfitting) and how was it solved?	The hardest problem was managing non-linear degradation near the end-of-life knee point where electrochemical capacity drops precipitously. Standard linear regression failed completely ($R^2 = 0.038$). I resolved this by engineering trailing window features (10/25-cycle SOH slopes) and tuning regularized XGBoost with tree depth constraints.
What new tool or library did I learn on my own?	I learned FastAPI for asynchronous ML serving with Pydantic validation, SHAP for game-theoretic feature attribution, Playwright for automated browser testing, and built an interactive React 18 dashboard with Tailwind CSS deployed live on Vercel and Render.
What would I do differently next time?	Next time, I would incorporate Physics-Informed Neural Networks (PINNs) to directly embed electrochemical partial differential equations into the model loss and implement real-time streaming CAN bus parsing on embedded microcontrollers.